Course Handout

Institute/School Name	Chitkara University Institute of Engineering and Technology		
Department Name	Department of Computer Science & Engineering		
Programme Name	Bachelor of Engineering (B.E.), Computer Science & Engineering		
Course Name	Linux Administration	Session	Jan – June 2026
Course Code	24CSE0213	Semester/Batch	4th /2024
L-T-P (Per Week)	0-0-6	Course Credits	03
Pre-requisite	Basics of Operating System	NHEQF Level[1]	5
Course Coordinator	Dr. Jatin Arora	SDG Number[4]	04,05,09,10,16

1. Objectives of the Course

Linux System Administration course is designed to help the student to become a Linux Admin Expert. The course is designed to shape the student as a Linux professional & help run applications, perform desired functions on system and networks, create a network configuration, and maintain security administration. The course provides a wide scope of learning and understanding of the subject. The objectives of the course are:

- To use Linux operating system knowledge for solving real world problem statements.
- To get familiar with the design, architecture and installation of Linux OS.
- To understand concepts of booting process, File system, working with files and directories, Editors and Filters/ Text processing commands of Linux OS.
- To understand basic concepts to manage the user, group of user's accounts on a system or on a network.
- To get familiar with shell scripting or program Linux system.

2. Course Learning Outcomes (CLOs)

Student should be able to:

	CLOs	Program Outcomes (PO)	NHEQF Level Descriptor[1]	No. of Lectures
CLO01	Understand fundamental concepts of Linux operating system.	PO1,PO2,PO4,PO5, PO9,PO12	Q1	18
CLO02	Apply concepts of Linux operating system in order to solve the real-life problems.	PO1,PO2,PO4,PO5, PO12	Q3	22
CLO03	Analyze the processes, file system and system directories in Linux operating system.	PO1,PO3,PO4,PO5, PO11,PO12	Q2	16
CLO04	Understand the working of Linux based system to manage the user or group of users in a network.	PO1,PO3,PO5, PO10,PO11	Q1	18
CLO05	Construct solutions for engineering problems by using shell script programming in Linux.	PO2,PO4,PO9, PO10,PO11	Q3	16
Total Contact Hours				90

CLO-PO Mapping

CLO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	Type of Assessment's[3]
CLO01		M			M			M					Summative
CLO02	M		M	M		M	M	M	M		M		Summative
CLO03						M	M		M				Summative
CLO04	M	M		M	M	M		H	H	M	H		Summative
CLO05	H	H	H	M	H		H	H			H		Summative

H=High, M=Medium, L=Low

3. Recommended Books:

B01: Linux Command Line and Shell Scripting Bible, Richard Blum, 4th edition, Wiley, 2021.
B02: Linux the Complete Reference, John Purcell, 7th edition, Walnut Creek, 1999.
B03: Your Unix - The Ultimate Guide, Sumitabha Das, 4th Edition, Tata McGraw-Hill, 2008.
B04: Linux Programming Bible, John Goerzen, 8th Edition, IDG Books, 2001.
B05: A Practical Guide to Linux, Mark G. Sobell, 3rd Edition by Pearson Education, 2013.
Reference Books:
B06: Unix Shell programming, Yashwant Kanetkar, 1st Edition, BPB Publications, 2004.
B07: Kubernetes - An Enterprise Guide, Scott Surovich, Marc Boorshtein 3rd Edition, Packt Publishing, 2020.

[1] NHEQF Level Descriptor, Refer to Annexure & Learning outcomes descriptors for qualification for all levels on the NHEQF
[3]Types of Assessments can be referred from Type of Assessments. Refer to Annexure.
[4]For SDG Mapping with Courses, Pl refer SDG Mapping policy for Courses

4. Other readings and relevant websites:

Serial No.	Link of Journals, Magazines, websites and Research Papers
1.	https://www.techtarget.com/searchdatacenter/definition/Linux-operating-system
2.	https://www.geeksforgeeks.org/introduction-to-linux-operating-system/
3.	https://resources.infosecinstitute.com/topic/installing-configuring-centos-7-virtualbox/
4.	https://ubuntu.com/tutorials/install-ubuntu-server#1-overview
5.	https://techlog360.com/basic-ubuntu-commands-terminal-shortcuts-linux-beginner/
6.	https://www.redhat.com/sysadmin/vim-commands
7.	https://learning.edx.org/course/course-v1:LinuxFoundationX+LFS101x+1T2017/home
8.	https://onlinecourses.swayam2.ac.in/aic20_sp24/course
9.	https://www.redhat.com/sysadmin/linux-command-basics-7-commands-process-management
10.	https://www.tutorialspoint.com/unix/unix-file-system.htm

5. Recommended Tools and Platforms

Oracle Virtual Box
Ubuntu Operating System

6. Course Plan: Theory+ Lab Plan

Lab Plan

Lecture Number	Topics
1-2	**Detail Discussion of Course Handout (CHO)**
3-4	Introduction to Linux: History and evolution of Linux, Differences between Linux distributions (Ubuntu flavors, Debian-based vs. others), Overview of Ubuntu and its derivatives (Ubuntu Desktop, Ubuntu Server, Kubuntu, Xubuntu, etc.)
5-8	Installation and Configuration: Preparing for Custom installation (hardware requirements, partitioning), Installing Ubuntu (Ubuntu Desktop and Server), Initial system setup (users, passwords, timezone, and basic configuration)
9-14	Basic Command Line Skills: Introduction to the terminal and shell (bash shell in Ubuntu), History, who, whoami, id, uname, help, man, cal, date, hostname, tty, clear, cat, touch, where is, what is. Navigating the file system (ls, cd, pwd, and Ubuntu directory structure), File and directory manipulation (cp, mv, rm, mkdir, rmdir)
15-18	Basic Command Line Skills: Viewing and editing files (cat, less, head, tail, nano, vim), File permissions and ownership (chmod, chown), Basic process management (ps, top, kill)
19-22	Package Management: Understanding package managers (APT on Ubuntu), Installing, updating, and removing software (apt, dpkg)
23-28	Basic Networking: Understanding IP addresses, DNS, and hostname, Basic network commands (ip, ping, netstat, traceroute), Configuring network interfaces (netplan, nmcli)
29-34	User and Group Management: Creating and managing users and groups (adduser, deluser, groupadd, usermod)
35-36	Configuring user permissions and access (chown, chmod), Understanding /etc/passwd and /etc/group files
ST-1 (Lecture 1-36)	
37-42	Streams: stdin, stdout, stderr, Redirection (>, >>, <) and piping: Sort, cut, paste
43-46	File Systems and Disk Management: Understanding file systems (ext4, xfs), Mounting and unmounting file systems (mount, umount), Disk partitioning and management tools (fdisk, parted, df, du)
47-50	Advanced File System Management: Understanding and managing file systems (LVM on Ubuntu), Backup and restore procedures (using rsync, tar), File system troubleshooting (fsck, dmesg)
51-56	Advanced Networking: Network configuration and management (netplan, nmcli), Firewall basics (ufw on Ubuntu), Network troubleshooting tools (tcpdump, wireshark)
57-60	System Monitoring and Performance Tuning: System monitoring tools (top, htop, iostat, vmstat), Performance tuning techniques (sysctl, tuned), Log management and analysis (syslog, journalctl)
61-64	Security Basics: Understanding Linux security principles, Managing firewalls (ufw), Implementing SSH and secure remote access (ssh, fail2ban), Security best practices
65-68	Understanding Shell Scripts, Executing and debugging shell scripts, Understanding shell variables, Special shell positional parameters, Reading in parameters, Parameter expansion in bash, Performing arithmetic in shell scripts.
69-72	Using programming constructs in shell scripts if . . . then, case command, for . . . do loop, while . . . do and until . . . do loops. Test Logic for Strings and numbers using test command, [ ], [[ ]].
ST-2 (Lecture 1-72)	

73-76	Advanced Shell Scripting & System Administration: Functions, functions taking parameters, return, exit status, function returning string.
77-82	Advanced string manipulation. Arrays: Simple arrays, adding element, deleting element, search and replace element
83-86	Task automation (cron, at, systemd services), Configuring and managing system services (systemctl)
87-90	Virtualization, Containers, and Advanced Security: Introduction to virtualization (KVM on Ubuntu), Working with containers (Docker, LXC/LXD on Ubuntu), Container orchestration basics (using microk8s for Kubernetes), Advanced firewall and security configurations (ufw, AppArmor on Ubuntu)
	End Term Exam (ETE-Complete Syllabus)

7. Delivery/Instructional Resources Lab Plan:

Lect. No.	Topics	CLO	Book No, CH No, Page No	TLM[2]	ALM[3]	Web References	Audio-Video
1-2	**Detail Discussion of Course Handout (CHO)**			Lecture Discussion			
3-4	Introduction to Linux: History and evolution of Linux, Differences between Linux distributions (Ubuntu flavors, Debian-based vs. others), Overview of Ubuntu and its derivatives (Ubuntu Desktop, Ubuntu Server, Kubuntu, Xubuntu, etc.)	1	B01-Chapter-1	Lecture Discussion	Quiz Challenge Assignment	https://www.redhat.com/en/topics/linux/what-is-linux	https://www.techtarget.com/searchdatacenter/definition/Linux-operating-system
5-8	Installation and Configuration: Preparing for Custom installation (hardware requirements, partitioning), Installing Ubuntu (Ubuntu Desktop and Server), Initial system setup (users, passwords, timezone, and basic configuration)	1	B01-Chapter-1 B02-Chapter-1	Lecture Discussion	Quiz Challenge Assignment	https://ubuntu.com/tutorials/how-to-run-ubuntu-desktop-on-a-virtual-machine-using-virtualbox#1-overview	https://www.youtube.com/watch?v=wSVA-VOwKgE
9-14	Basic Command Line Skills: Introduction to the terminal and shell (bash shell in Ubuntu), History, who, whoami, id, uname, help, man, cal, date, hostname, tty, clear, cat, touch, where is, what is. Navigating the file system (ls, cd, pwd, and Ubuntu directory structure), File and directory manipulation (cp, mv, rm, mkdir, rmdir)	2	B01-Chapter-2 B03-Chapter-2, 3	Lecture Discussion	Quiz Challenge Assignment	https://www.edureka.co/blog/linux-commands/	https://onlinecourses.swayam2.ac.in/aic20_sp24/announcements?force=true
15-18	Basic Command Line Skills: Viewing and editing files (cat, less, head, tail, nano, vim), File permissions and ownership (chmod, chown), Basic process management (ps, top, kill)	3,4	B01-Chapter-3, 4, 10 B03-Chapter-4, 10	Lecture Discussion	Quiz Challenge Assignment	https://www.tutorialspoint.com/top-5-best-linux-text-editors	https://ru.coursera.org/lecture/linux-fundamentals/editing-text-files-xkv0S
19-22	Package Management: Understanding package managers (APT on Ubuntu), Installing, updating, and removing software (apt, dpkg)	2,3	B01-Chapter-9	Lecture Discussion	Quiz Challenge Assignment	https://ubuntu.com/server/docs/package-management	https://www.youtube.com/watch?v=yxc2ntmH9xY

[2] Teaching Learning Methods, Refer to Annexure
[3] Active Learning Methods

23-28	Basic Networking: Understanding IP addresses, DNS, and hostname, Basic network commands (ip, ping, netstat, traceroute), Configuring network interfaces (netplan, nmcli)	4	B03-Chapter-11	Lecture Discussion	Quiz Challenge Assignment	https://learning.edx.org/course/course-v1:LinuxFoundationX+LFS101x+1T2017/home	https://onlinecourses.swayam2.ac.in/aic20_sp24/course
29-34	User and Group Management: Creating and managing users and groups (adduser, deluser, groupadd, usermod)	3, 4	B01-Chapter-7	Lecture Discussion	Quiz Challenge Assignment	https://docs.fedoraproject.org/en-US/fedora/latest/system-administrators-guide/basic-system-configuration/Managing_Users_and_Groups/	https://www.youtube.com/watch?v=FtwRe8w2kWI
35-36	Configuring user permissions and access (chown, chmod), Understanding /etc/passwd and /etc/group files	4	B01-Chapter-7	Lecture Discussion	Quiz Challenge Assignment	https://learning.edx.org/course/course-v1:LinuxFoundationX+LFS101x+1T2017/home	https://onlinecourses.swayam2.ac.in/aic20_sp24/course
37-42	Streams: stdin, stdout, stderr, Redirection (>, >>, <) and piping: Sort, cut, paste	2, 3	B01-Chapter-15 B03-Chapter-9	Lecture Discussion	Quiz Challenge Assignment	https://spoken-tutorial.org/watch/Linux/Redirection+Pipes/English/ https://spoken-tutorial.org/watch/Linux/Simple+filters/English/	https://www.youtube.com/channel/UCcLQJOfR-MCcI5RtIHFl6Ww https://www.youtube.com/watch?v=WH9HUEKtEzQ&t=35s
43-46	File Systems and Disk Management: Understanding file systems (ext4, xfs), Mounting and unmounting file systems (mount, umount), Disk partitioning and management tools (fdisk, parted, df, du)	3	B01-Chapter-8 B03-Chapter-21	Lecture Discussion	Quiz Challenge Assignment	https://help.ubuntu.com/stable/ubuntu-help/disk-partitions.html.en	https://www.youtube.com/watch?v=AQ8u-Kx_MSQ&list=PLbvUFWvKlQnnej_Vu588Eor8nc4ENrA0i
47-50	Advanced File System Management: Understanding and managing file systems (LVM on Ubuntu), Backup and restore procedures (using rsync, tar), File system troubleshooting (fsck, dmesg)	3	B01-Chapter-8 B03-Chapter-21	Lecture Discussion	Quiz Challenge Assignment	https://www.geeksforgeeks.org/tar-command-linux-examples/	https://onlinecourses.swayam2.ac.in/aic20_sp24/announcements?force=true
51-56	Advanced Networking: Network configuration and management (netplan, nmcli), Firewall basics (ufw on Ubuntu), Network troubleshooting tools (tcpdump, wireshark)	4	B03-Chapter-23	Lecture Discussion	Quiz Challenge Assignment	https://learning.edx.org/course/course-v1:LinuxFoundationX+LFS101x+1T2017/home	https://onlinecourses.swayam2.ac.in/aic20_sp24/course
57-60	System Monitoring and Performance Tuning: System monitoring tools (top, htop, iostat, vmstat), Performance tuning techniques (sysctl, tuned), Log management and analysis (syslog, journalctl)	2	B01-Chapter-16	Lecture Discussion	Quiz Challenge Assignment	https://www.fosslinux.com/134995/how-to-use-sysctl-for-effective-kernel-tuning-in-ubuntu.htm	https://www.youtube.com/watch?v=Kzpm-rGAXos
61-64	Security Basics: Understanding Linux security principles, Managing firewalls (ufw), Implementing SSH and secure remote access (ssh, fail2ban), Security best practices	2	B03-Chapter-11	Lecture Discussion	Quiz Challenge Assignment	https://www.hostinger.in/tutorials/how-to-configure-firewall-on-ubuntu-using-ufw/	https://www.youtube.com/watch?v=XtRXm4FFK7Q

65-68	Understanding Shell Scripts, Executing and debugging shell scripts, Understanding shell variables, Special shell positional parameters, Reading in parameters, Parameter expansion in bash, Performing arithmetic in shell scripts,	5	B01-Chapter-11, 12	Lecture Discussion	Quiz Challenge Assignment	www.redhat.com/academy	www.redhat.com/academy
69-72	Using programming constructs in shell scripts if . . . then, case command, for . . . do loop, while . . . do and until . . . do loops. Test Logic for Strings and numbers using test command, [ ], [[ ]].	5	B01-Chapter-13 B03-Chapter-18	Lecture Discussion	Quiz Challenge Assignment	www.redhat.com/academy	www.redhat.com/academy
73-76	Advanced Shell Scripting & System Administration: Functions, functions taking parameters, return, exit status, function returning string.	4, 5	B01-Chapter-17	Lecture Discussion	Quiz Challenge Assignment	https://tldp.org/LDP/abs/html/	https://www.youtube.com/watch?v=cQepf9fY6cE&list=PLS1QulWo1RIYmaxcEqw5JhK3b-6rgdWO_
77-82	Advanced string manipulation. Arrays: Simple arrays, adding element, deleting element, search and replace element	3, 5	B01-Chapter-17	Lecture Discussion	Quiz Challenge Assignment	https://tldp.org/LDP/abs/html/	https://www.youtube.com/watch?v=cQepf9fY6cE&list=PLS1QulWo1RIYmaxcEqw5JhK3b-6rgdWO_
83-86	Task automation (cron, at, systemd services), Configuring and managing system services (systemctl)	3, 5	B01-Chapter-16	Lecture Discussion	Quiz Challenge Assignment	www.redhat.com/academy	www.redhat.com/academy
87-90	Virtualization, Containers, and Advanced Security: Introduction to virtualization (KVM on Ubuntu), Working with containers (Docker, LXC/LXD on Ubuntu), Container orchestration basics (using microk8s for Kubernetes), Advanced firewall and security configurations (ufw, AppArmor on Ubuntu)	2, 4	B07-Chapter-2	Lecture Discussion	Quiz Challenge Assignment	https://phoenixnap.com/kb/install-kubernetes-on-ubuntu	https://www.youtube.com/watch?v=briu_Ev89sw

8. Remedial Classes[4]

Slow Learners	Average Learners	Fast Learners
• Remedial Classes on Saturdays • Encouragement for improvement using Peer Tutoring • Use of Audio and Visual Materials • Use of Real-Life Examples	• Workshops • Formative Exercises used to highlight concepts and notions • E-notes and E-exercises to read ahead of the pedagogic material.	• Engaging students to hold hands of slow learners by creating a Peer Tutoring Group • Design solutions for complex problems • Design solutions for complex problems • Presentation on topics beyond those covered in CHO

9. Self-Learning[5]

Assignments to promote self-learning, survey of contents from multiple sources.

S.No	Assignment Topic	CLO	ALM	References / MOOCs
1	Study Linux evolution and compare at least 4 distributions(Ubuntu, Fedora, Arch, CentOS) on package management, use cases, and community support. Submit a comparison matrix.	1	Survey + Contradiction Analysis	https://www.redhat.com/en/topics/linux/what-is-linux
2	Study Linux filesystem hierarchy (/etc, /proc, /var, /usr). Map purpose + real configuration examples.	3	Concept	https://www.edureka.co/blog/linux-commands/
3	Compare traditional permissions vs ACLs. Include real commands and use cases.	3	Exploratory Learning	www.redhat.com/academy

[4] Refer to Annexure
[5] Refer to Annexure

4	Survey APT, YUM/DNF, Compare speed, security, rollback features.	2	Survey	www.redhat.com/academy
5	Compare ip, ifconfig, netstat, ss, nmcli. Explain why modern Linux prefers newer tools.	2	Models	https://learning.edx.org/course/course-v1:LinuxFoundationX+LFS101x+1T2017/home
6	Study enterprise user management practices in Linux. Suggest security guidelines for lab systems.	4	Case Study	User and Group Management: Creating and managing users and groups (adduser, deluser, groupadd, usermod) https://docs.fedoraproject.org/en-US/fedora/latest/system-administrators-guide/basic-system-configuration/Managing_Users_and_Groups/
7	Analyze boot stages (BIOS/UEFI → GRUB → Kernel → systemd). Include flow diagram.	1	Analytical Learning	https://ubuntu.com/tutorials/how-to-run-ubuntu-desktop-on-a-virtual-machine-using-virtualbox#1-overview
8	Survey real-world shell scripts used for backups, monitoring, log cleanup. Explain logic (no coding mandatory).	5	Case Study	https://tldp.org/LDP/abs/html/
9	Compare cron jobs and systemd timers with advantages, limitations, and use cases.	5	Analytical Learning	www.redhat.com/academy
10	Compare KVM, Docker, LXC with real-world deployment examples.	2	Contradiction Analysis	https://phoenixnap.com/kb/install-kubernetes-on-ubuntu

10. Delivery Details of Content Beyond Syllabus[6]

Content beyond syllabus covered (if any) should be delivered to all students that would be planned, and schedule notified accordingly.

S.No	Advanced Topics, Additional Reading, Research papers and any	CLO	POs	ALM	References/MOOCS
1	Deep dive into nice, renice, ulimit, cgroups and real-world performance tuning concepts.	NA	1, 3, 4, 5	Problem-Based Learning	www.redhat.com/academy

11. Evaluation Scheme & Components:

Assessment Type[7]	Evaluation Component[8]	Type of Component[9]	No. of Assessments[10]	% Weightage of Component	Max. Marks	Mode of Assessment	CLO
Formative	Component1	Formative Assessment	02*	20%	20	--	1 - 5
Summative	Component2	Sessional Tests (STs)	02*	40%	40	Computer based online in Campus	1 – 4
Summative	Component3	End Term Examination	01**	40%	40	Computer based online in Campus	1 - 5
	Total		**100%**				

* No makeup exam will be conducted for any of the Formative Assessments (FAs) and Sessional Tests (STs).
* For both FAs and STs, the best score out of the two will be considered for the final marks computation.
**As per Academic Guidelines, a minimum of 75% attendance is required to become eligible for appearing in the End Semester Examination.

[6] Refer to Annexure
[7] Refer to Annexure 2 of NCrF
[8] Refer to Annexure
[9] Refer to Annexure
[10] Refer to Annexure

12. Syllabus of the Course:

S. No.	Topic (s)	No. of Lectures	Weightage %
1	**Detail Discussion of Course Handout (CHO)** Introduction to Linux: History and evolution of Linux, Differences between Linux distributions (Ubuntu flavors, Debian-based vs. others), Overview of Ubuntu and its derivatives (Ubuntu Desktop, Ubuntu Server, Kubuntu, Xubuntu, etc.) Installation and Configuration: Preparing for Custom installation (hardware requirements, partitioning), Installing Ubuntu (Ubuntu Desktop and Server), Initial system setup (users, passwords, timezone, and basic configuration) Basic Command Line Skills: Introduction to the terminal and shell (bash shell in Ubuntu), History, who, whoami, id, uname, help, man, cal, date, hostname, tty, clear, cat, touch, where is, what is. Navigating the file system (ls, cd, pwd, and Ubuntu directory structure), File and directory manipulation (cp, mv, rm, mkdir, rmdir) Basic Command Line Skills: Viewing and editing files (cat, less, head, tail, nano, vim), File permissions and ownership (chmod, chown), Basic process management (ps, top, kill) Package Management: Understanding package managers (APT on Ubuntu), Installing, updating, and removing software (apt, dpkg) Basic Networking: Understanding IP addresses, DNS, and hostname, Basic network commands (ip, ping, netstat, traceroute), Configuring network interfaces (netplan, nmcli) User and Group Management: Creating and managing users and groups (adduser, deluser, groupadd, usermod) Configuring user permissions and access (chown, chmod), Understanding /etc/passwd and /etc/group files	**36**	**40%**
ST-1			
2	Streams: stdin, stdout, stderr, Redirection (>, >>, <) and piping: Sort, cut, paste File Systems and Disk Management: Understanding file systems (ext4, xfs), Mounting and unmounting file systems (mount, umount), Disk partitioning and management tools (fdisk, parted, df, du). Advanced File System Management: Understanding and managing file systems (LVM on Ubuntu), Backup and restore procedures (using rsync, tar), File system troubleshooting (fsck, dmesg). Advanced Networking: Network configuration and management (netplan, nmcli), Firewall basics (ufw on Ubuntu), Network troubleshooting tools (tcpdump, wireshark). System Monitoring and Performance Tuning: System monitoring tools (top, htop, iostat, vmstat), Performance tuning techniques (sysctl, tuned), Log management and analysis (syslog, journalctl) Security Basics: Understanding Linux security principles, Managing firewalls (ufw), Implementing SSH and secure remote access (ssh, fail2ban), Security best practices. Understanding Shell Scripts, Executing and debugging shell scripts, Understanding shell variables, Special shell positional parameters, Reading in parameters, Parameter expansion in bash, Performing arithmetic in shell scripts, Using programming constructs in shell scripts if . . . then, case command, for . . . do loop, while . . . do and until . . . do loops. Test Logic for Strings and numbers using test command, [ ], [[ ]].	**36**	**80%**
ST-2			
3	Advanced Shell Scripting & System Administration: Functions, functions taking parameters, return, exit status, function returning string. Advanced string manipulation. Arrays: Simple arrays, adding element, deleting element, search and replace element. Task automation (cron, at, systemd services), Configuring and managing system services (systemctl) Virtualization, Containers, and Advanced Security: Introduction to virtualization (KVM on Ubuntu), Working with containers (Docker, LXC/LXD on Ubuntu), Container orchestration basics (using microk8s for Kubernetes), Advanced firewall and security configurations (ufw, AppArmor on Ubuntu)	**18**	**100%**
End Term Examination (ETE-Complete Syllabus)			

13. Academic Integrity Policy:

Education at Chitkara University builds on the principle that excellence requires freedom where Honesty and integrity are its prerequisites. Academic honesty in the advancement of knowledge requires that all students and Faculty respect the integrity of one another's work and recognize the importance of acknowledging and safeguarding intellectual property. Any breach of the same will be tantamount to severe academic penalties.

This Document is approved by:

Designation	Name	Signature
Course Coordinator	Dr. Jatin Arora	
Head-Academic Delivery	Dr. Mrinal Paliwal	
Dean	Dr. Rishu Chhabra	

Date(DD/MM/YYYY)	16.12.2025	

Annexure

1. **Pre- requisite**
   Mention The Pre-requisite skill set or course/s if it is expected to be studies before this course, otherwise write "not applicable".

2. **NHEQF levels**

The NHEQF levels represent a series of sequential stages expressed in terms of a range of learning outcomes against which typical qualifications are positioned/located. NHEQF level 4.5 represents learning outcomes appropriate to the first year (first two semesters) of the undergraduate programme of study, while Level 8 represents learning outcomes appropriate to the doctoral-level programme of study.

Table 1: Higher education qualifications at different levels on the NHEQF

NHEQF level	Examples of higher education qualifications located within each level
Level 4.5	Undergraduate Certificate. Programme duration: First year (first two semesters) of theundergraduate programme, followed by an exit 4-credit skills-enhancement course(s).
Level 5	Undergraduate Diploma. Programme duration: First two years (first four semesters) of the undergraduate programme, followed by an exit 4-credit skills-enhancement course(s) lasting two months.
Level 5.5	Bachelor's Degree. Programme duration: First three years (Six semesters) of the four-yearundergraduate programme.
Level 6	Bachelor's Degree (Honours/ Honours with Research). Programme duration: Four years(eight semesters).
Level 6	Post-Graduate Diploma. Programme duration: One year (two semesters) for those whoexit after successful completion of the first year (two semesters) of the 2-year master's programme.
Level 6.5	Master's degree. (e.g. M.A., M.Com., M.Sc., etc.) Programme duration: Two years (four semesters) after obtaining a 3- year Bachelor's degree (e.g. B.A., B.Sc., B.Com.etc.).
Level 6.5	Master's degree. (e.g. M.A., M.Com., M.Sc., etc.) Programme duration: One year (two semesters) after obtaining a 4 - year Bachelor's degree (Honours/ Honours with Research) (e.g. B.A., B.Sc., B.Com. etc.).
Level 7	Master's degree.(e.g. M.E./M.Tech. etc.) Programme duration: Two years (four semesters) after obtaining a 4-year Bachelor's degree. (e.g. B.E./B.Tech. etc.)
Level 8	Doctoral Degree

3. **NHEQF level descriptors**

Each NHEQF level is structured based on the defined learning outcomes which lead to the expected graduate attributes/profile. The level descriptors reflect the expected outcomes of learning that should be achieved and demonstrated by graduates of a specific programme of study leading to a qualification at a specific NHEQF level.

Click Learning outcomes descriptors for qualification for all levels on the NHEQF

4. **Course Outcomes**
   The number of Course Outcomes is recommended to be 4-5 for courses that do not contain practical component and 6 for those courses with a practical component. Flexibility can be sought by the post-graduate courses in this regard.

5. **Theory/lab Plan**
   The following are the guidelines to be followed while creating plans

- Each session may be planned for a duration of 45/50mins (irrespective of the double hour or single hour scheduled in timetable).
- Every session must incorporate at least one active learning method which may or may not be part of the assessments.
- Put BoS Approved Syllabus in the topics. Deviations (if any) from BoS approved syllabus must be brought to the notice of BoS chairman & Dean Academics, After approval, revised handout should be submitted.
- The Topics elaborated in the Theory/Lab plan must match those in the course execution plan.

6. **Teaching Learning Methods**
   The following are some of the Teaching & Learning methods that can be incorporated in session wise teaching learning plan.
   - **Teacher-centered Learning Methods:**
     i. Lecture
     ii. Discussion
     iii. Demonstration method using a simulation or a tool
     iv. Reviewing
     v. Questioning

   - **Learner-centered teaching & Learning methods:**

i. **Active learning**, in which students solve problems, answer questions, formulate questions of their own, discuss, explain, debate, or brainstorm during class;
ii. **Cooperative learning**, in which students work in teams on problems and projects under conditions that assure both positive interdependence and individual accountability; and
iii. **Inductive teaching and learning**, in which students are first presented with challenges (questions or problems) and learn the course material in the context of addressing the challenges.
iv. **Inductive methods** include inquiry-based learning, case based instruction, problem-based learning, project-based learning, discovery learning, and just-in-time teaching. It is important to integrate authentic, reflective and collaborative learning experiences when designing for student-centered learning.

7. **Active Learning Methods**

The following are some of the Active Learning Methods that can be incorporated in session wise teaching learning plan.

- One Minute Paper
- Group Discussion
- Student-Created PPT, Charts, Matrices, Flowcharts, Models
- The Fish Bowl
- Debate
- Video Synthesis
- Quiz/Test Questions
- Brain Storming Sessions
- Case Study
- Shadowing
- Leading Question
- Puzzle, Enigma, Contradiction
- Statement-Opinion-Summary
- Think / Pair / Share
- Peer Review
- Just in Time Teaching
- Statement-Opinion-Summary
- Peer Survey
- Focused Listing
- Role-Playing
- Student Field Work with Reflection
- Infusing Humor into Class Sessions
- Inviting Effective Guest Speakers

8. Remedial Classes

After every Sessional Test, identify weak learners, provide supplement course handout. Student list and Impact Observed report should be submitted to Dean through proper channel.

9. Self Learning

Plan 10% of topics in self-learning mode with discussions, ALM's and Assessment happing in the class.

10. Content Beyond Syllabus

Plan Advanced Topics, Experiments, Additional Reading, Research papers in self-learning mode with ALM's and Assessment happing in the regular class or lab. Usually caters advanced learners. Identify Advanced learners. For Extra classes, schedule should be notified accordingly.

11. Assessment Type

1. Assessment broadly can be classified into the following types:

a. **Diagnostic assessments**: Diagnostic assessments are intended to help teachers identify what students know and can do in different domains to support their students' learning. These help teachers determine strengths of students in various areas to
better address their specific needs.
b. **Formative assessments**: Formative assessment refers to a wide variety of methods that teachers use to conduct in-process evaluations of student comprehension, learning needs, and academic progress during a lesson, unit, or a course. Formative assessments help teachers identify concepts that students are struggling to understand, skills they are having difficulty acquiring, or learning standards they have not yet achieved so that adjustments can be made to lessons, instructional techniques, and academic support.
c. **Summative assessments**: Summative assessment is an assessment administered at the end of an instructional unit in a course. These assessments are intended to evaluate student learning by comparing performance to a standard or benchmark.
d. **Ipsative assessments**: Ipsative assessment involves comparisons between past and current work to identify a learner's growth over time, rather than progress toward an external set of criteria. Therefore, Ipsative assessment is an internal or self-referenced assessment.
e. **Norm-referenced assessments**: Norm-referenced tests report whether test takers performed better or worse than a hypothetical average student, which is determined by comparing scores against the performance results of a statistically selected group of test takers, typically of the same age or grade level, who have already taken the exam.
f. **Criterion-referenced assessments**: Criterion-Reference tests measure the performance of test takers against the criteria covered in the curriculum.
g. **Peer-to-Peer randomised Assessments**: Peers will be able to provide assessment in this case
h. **Industry Validation of Effectiveness**: In the Vocation Education, Industry validation of effectiveness of training is particularly important.
i. **Self-assessments**: To evaluate how much the learner has grasped by self-learning.

2. Other Assessment Methods: Conducting an assessment takes time, thought, attention, planning, and often collaboration. Each assessment tool, whether a short survey or detailed rubric, will be useful only insofar as it both addresses the outcomes well and is feasible to use.

a**. Rubrics**: For assessing qualitative student work such as essays, projects, reports, or presentations. Rubrics serve well to clearly denote the

specific expectations for an assignment, for collecting data for assessment of student learning outcomes. and for student performance. Rubrics can be used for grading, for providing feedback to students, and for informing and encouraging students to think about their own learning.
b. **Portfolios and E-Portfolio**: Portfolios can provide a window into the process of student learning across a semester-long project that can be assessed (usually by using a rubric).
c. **Curriculum Mapping**: A good curriculum map can serve to focus assessment, and the improvements that follow, where it will be most useful, informative, or effective.
d. **Structured Interviews**: While time-consuming, structured interviews are useful when specific questions need to be asked. It also leaves room for unplanned topics or ideas to emerge.
e. **Student Experience Surveys**: Student experience in research universities (SERU), including administration of on-line census SERU Undergraduate and Graduate Surveys, can yield important information about student perceptions and experiences.

12. Evaluation Component & Types

As per LMs we need to figure it out whether it is component 1, 2 or 3. In Types of Evaluation Component, we need to specify what type of evaluation we are performing like Continuous Evaluation or Sessional Test or End Term Examination.

13. No. of Assessments and Weightage of Components

Department will give guideline for number of assessments, mandatory or optional and weightage.